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CS 2300 – Tues/Thurs (12:15pm)

Prof. Pam Russell

PA 3 – Deliverables

Question #1:

Each block of plaintext must be represented as a 4x1 column vector to match dimensions of the 4x4 encryption matrix. This will let the matrix multiplication work and will ensure that the encryption is performed correctly.

Pseudocode:

1. Initialize and define encoding matrix (A)
 - Create a 4x4 matrix A
2. Read plaintext from the input file
 - Open file "input-A21-1.txt" for reading
 - Read contents, remove extra spaces
3. Convert plaintext to ASCII values
 - For each character in plaintext
 1. convert it to corresponding ASCII value
 2. Store ASCII values in a list "ascii_vals"
4. Add zeros to the ASCII values list
 - Check if length of ascii_vals list is a multiple of 4
 1. If not, add zeros to list until length is divisible by 4.
5. Reshape and transpose the ASCII values
 - Reshape ascii_vals list into matrix B w/ 4 rows
 - Transpose matrix B to ensure it aligns with encryption matrix for multiplication
6. Display plaintext matrix (B)
 - Print matrix B to show reshaped plaintext
7. Encrypt plaintext using the encoding matrix A
 - Perform matrix multiplication ($C = A * B$)
 1. C is encrypted ciphertext
 - Flatten result matrix C in column based order to get 1D list of ciphertext values
8. Write the ciphertext to output file
 - Open file "input-A22-fixed.txt" to write
 - Write ciphertext values as a space-separated string into file
9. Read ciphertext back from output file
 - Open file "input-A22-fixed.txt" for reading
 - Read ciphertext
 1. convert values into integers

1. store them in a list `c_vals`.
10. Reshape ciphertext back into matrix `C`
 - Reshape list `c_vals` into matrix `C` w/ 4 rows
 - Print reshaped matrix `C`.
11. Decrypt the ciphertext
 - Compute matrix `A` inverse (`A_inv`).
 - Perform matrix multiplication to decrypt ciphertext
 1. `B_decoded = A_inv * C`.
 2. Round result of decryption and convert values to ints
12. Transpose decoded matrix
 - Transpose matrix `B_decoded` to restore original character order
13. Convert ASCII vals back to characters
 - For each value in `B_decoded`
 1. convert it back to its respective character w/ `chr()` function.
 2. Filter out values which aren't valid ASCII characters
14. Reconstruct decrypted message
 - Join characters to form final decrypted message
15. Display the decrypted message
 - Print decrypted message